

TOM / ATOMOS

Unified operator specification

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Consolidation U1 | 20 September 2026

01 / SELECTED ROUTE

Exact operator core; demand-driven readouts

A signed-field operator system derives candidates from available evidence, qualifies publication by forward time, preserves committed definitions, and constructs application readouts only when requested. A coordinate grid, metric, graph, physical carrier and hardware backend are not mandatory parts of the definition.

Source priority. TOM v0.7.2 supplies the semantic core **[A]**. The second document supplies the overlapping binary gate, finite-capacity, lookup and just-in-time requirements **[B]**. This consolidation's explicit reconciliation choices are marked **[U]**; they are not attributed retroactively to either source. [A pp. 1–6; B pp. 8–12, 35–39]

Retained defining expression

$\text{InverseOccam}[\text{Arch}[C[J[T]]; E[C[J[T]]; \text{InvLog}[\phi]]]]$

qualified by forward_T_precedence

Shared requirement	Unified literal meaning
Every role is SDF-qualified	Every admitted role or compound has a typed signed-field term, symbolic obligations, or an explicit rejection.
Operator precedence	Ordered operands determine derivation; forward temporal qualification separately determines publication.
One-bit participation	A binary gate selects passive evaluation or an eligible candidate. It is not the field sign or a stored history.
Finite execution	Declared representation, input, working-memory and history budgets limit admission. Failure publishes no partial state.
One-dimensional timeline	Commit order is forward-only. Historical selection is read-only and limited to retained, reconstructible records.
No compulsory coordinates	Geometry, topology, graphs and lookup tables are application profiles, not replacements for the operator expression.

[A pp. 2–6; B pp. 8–12, 15, 21–22, 37–39; U]

Optimization decision [U]. Use one exact, provenance-preserving operator core, share repeated derivations, keep readouts disposable, and add only required adapters. This minimizes mandatory state and unsupported mechanisms under the source's non-erasure and temporal constraints. Neither PDF supplies comparative benchmarks establishing a globally fastest design.

02 / CORE CONTRACT

One representation; one atomic transition

Universal signed-field term

$$F = (v, s, r, \mathbf{o}, u, n, a, p, h; \Omega).$$

Here v is an integer, reduced rational, exact $\mathbb{Z}[\phi]$ value or symbolic expression; $s \in \{-, 0, +, ?\}$ is its declared sign; r is the role; $\mathbf{o}$ contains ordered operands; u is a unit tag; n is the evaluation tick; a is the latest evidence-availability tick; p preserves provenance; h references committed history. Ω explicitly records unresolved law obligations. Unknown is not zero. A zero readout does not erase its operands.

[A pp. 3–5; U: explicit Ω]

Registry role	Contract retained in U1
$T, J[T]$	Read temporal state and available evidence; apply only a named jitter law.
C, E, Arch	Derive the causal term; construct its ordered echo with $\kappa = -\ln \phi$; preserve direct and echo operands in the readout. Missing scalar laws stay open.
InverseOccam	Preserve the defining term, provenance and committed prefix. Repeated qualification does not erase or duplicate the definition.
TOM, A, Klein, Matryoshka	Retain literal declarations. A is not silently aliased to T ; topology and nesting laws are not guessed.
SELF, Later_T	Retain deferred terms and explicit guards; reject an unavailable or unguarded continuation.
Observation, Actualization	Preserve input value, availability, units, law identity and admission outcome. Actualization requires its application law.

[A pp. 4–7]

Candidate derivation

Read one committed snapshot S and evidence e available at tick n . Evaluate $c = C[J[T]]$ once, then $\epsilon = E[c; \kappa]$, $r = \text{Arch}[c; \epsilon]$ and $q = \text{InverseOccam}[r]$. Sharing c preserves both source operand references; it does not replace them with equal scalar values. The printed echo profile is $E[c; \kappa] = \kappa c$; this echo is not thermodynamic entropy. [A pp. 5–6; U: evaluation sharing]

Publication gate [U]

A candidate may commit only when $g = 1$, $n > n_{\text{last}}$, all evidence has availability $\leq n$, required guards and laws pass, the prefix is intact, and all exact representations fit the declared budgets. Symbolic terms may commit as symbolic terms; an unresolved law cannot authorize a concrete application result.

$$(S', W') = \begin{cases} (\text{publish}(q, n), W \parallel (n, q, e)), & \text{all checks pass,} \\ (S, W), & \text{otherwise.} \end{cases}$$

The operator returns a term or an explicit status, including PROGRAM_ERROR, FUTURE_EVIDENCE, TIME_ORDER, UNBOUND_LAW, INVARIANT_FAIL or CAPACITY. A declared check order makes multiple failures deterministic. [A pp. 3, 6, 8–9; U]

03 / EXPLICIT RECONCILIATIONS

Lowercase phi, binary state and time

Exact lowercase phi [U]

U1 selects the positive lower ratio used in B, without decimal arithmetic:

$$0 < \phi < 1, \quad \phi^2 = 1 - \phi, \quad \kappa = -\ln \phi > 0.$$

Store $a + b\phi$ as $(a, b) \in \mathbb{Z}^2$. Then

$$\phi(a + b\phi) = b + (a - b)\phi,$$

$$(a, b)(c, d) = (ac + bd, ad + bc - bd).$$

κ remains symbolic. Unit tags do not turn the algebra into a spatial metric. [A pp. 2-5; B pp. 1, 7, 9; U: reconciled algebra]

Source conflict, not a silent correction. A, page 4, prints multiplication by phi as $(a, b) \mapsto (b, a + b)$. That law implies $\phi^2 = 1 + \phi$, not the lower-ratio law above. Preserve it only as a separately named legacy rule. For its positive root $\phi_+ = 1 + \phi$,

$$a + b\phi_+ = (a + b) + b\phi.$$

This changes the value basis, not the meaning of the multiplier. U1 does **not** assert unchanged v0.7.1/v0.7.2 recurrence vectors or binary compatibility after selecting the lower-ratio rule.

One bit is one bit [U]

Use $g \in \{0, 1\}$ for B's AB gate: $g = 0$ permits passive/read-only evaluation; $g = 1$ permits a candidate to seek admission. B's four-value notation $\{00, 01, 10, 11\}$ on page 9 is not retained as a one-bit type. Field sign, optional orientation state, presence detection and the timeline are separate values. [B pp. 8-11, 33-35]

Jitter is not the clock or identity

The reference jitter law reported in A is $J(n) = (-1)^{\text{popcount}(n)}$ for nonnegative integer n . Retaining that named law does not identify it with B's double sinusoid. A two-hertz carrier remains an optional input profile, not a universal biological identifier or a temporal axiom. [A p. 4; B pp. 16-18; U]

A clock profile must specify whether two hertz means two complete cycles or two toggle events per second. Two toggle events per second give one complete binary cycle per second. Neither interpretation supplies the elapsed-time unit of T without a declared mapping. [B pp. 16-17; U]

Forward commit; read-only historical selection [U]

Let $W = ((n_i, F_i, e_i))_{i=0}^k$ denote the retained committed sequence. $T_{\text{read}}(\tau)$ selects the latest retained record with $n_i \leq \tau$ within the available retained interval; otherwise it returns OUT_OF_RANGE. It does not interpolate missing observations, rewrite the past or disclose future evidence. Predictions require a separate model and remain predictions. [A pp. 4, 6-7; B pp. 21-22, 37-39; U]

Keep $\text{SDF}_t = t - t_0$ separate from C 's stated future-negative/past-positive orientation; each application declares the sign mapping. [A p. 5]

04 / MEMORY AND LOOKUP

Preserve evidence; discard derived workspace**Resolve non-erasure versus “novelty is not stored” [U]**

The authoritative state is the committed operator history plus the source and evidence needed to reproduce it. Candidate scratch space, temporary coordinates and recomputable readout caches are disposable. An accepted observation or defining term is not disposable merely because its value is novel.

[A pp. 2–3, 6; B pp. 21–25, 38–39]

The retained meaning of “no novelty storage” is narrower: unfamiliar inputs do not automatically create new operators, laws or permanent geometry. An unmatched input may return `OUT_OF_PROFILE`; an accepted symbolic input retains its provenance. This resolves the conflict explicitly rather than claiming both total erasure and exact historical recovery.

Finite capacity is an admission rule

Declare limits for exact-value limbs, term records, source/evidence bytes, history, indexes and transient workspace. Reserve the complete candidate’s required capacity before publication. Exhaustion returns `CAPACITY` and leaves the committed prefix unchanged. With fixed finite storage and non-erasure, execution can stop; it cannot promise indefinitely growing history. [A pp. 3, 6, 8; B pp. 2–3; U]

This replaces B’s physical “deadlock” claim with a precise computational rule: **no admitted update beyond the budget**. It does not imply that topology stops physical motion or that every occupied graph prohibits every state transition. [B pp. 5, 12, 32–33; U]

Demand-driven readout and finite lookup [U]

For a declared profile P , form a readout only on request:

$$R_{P,\tau} = D_P(T_{\text{read}}(\tau)).$$

A cache or lookup table may accelerate D_P only when its exact key includes all semantically relevant inputs: role and operands, registry/profile version, algebra basis, temporal state and evidence identity. A hit returns the same term or status as the uncached evaluator. A miss invokes that evaluator or returns a declared rejection; it never manufactures a nearest answer. [A pp. 3, 8; B pp. 37–39; U]

A log-polar or spherical lookup format is optional. It requires a declared domain, radial/phase interpretation, exact bin-assignment rule, error bounds where quantized, and capacity. Naming a table “log-polar” does not supply those laws or eliminate key construction, evidence ingestion, exact arithmetic or output cost. [B pp. 2–3, 20, 26; U]

Selected optimizations [U]

Store source bytes once with immutable references; share repeated evaluations of identical full inputs; retain defining operands even when readouts coincide; compute geometry only when a bound readout asks for it. No full-space rasterization, all-pairs graph or precomputed global route table is required by the core.

Historical replay needs retained records and evidence, or an explicitly specified generator that reproduces the requested information. Discarded observations do not become recoverable by moving a time-line selector. Cache eviction and authoritative-history deletion are different operations.

05 / APPLICATION BOUNDARY

Keep the names; bind the mechanics explicitly

All following roles remain eligible signed-field operators. Their application laws are optional and unresolved until declared; they do not become core axioms through their names. [A pp. 4–5, 7, 9–10; U]

Role or profile	Literal contract selected for U1
DoublePack(F, K); Klein	Retain the field term and topology qualifier together, with both definitions and obligations. A numerical topology profile must supply domain, identifications and compatible sign/orientation laws. Multiplying a field by a sign is not accepted as that specification.
Primitive and growth fields	PyramidSide, Cone, Sphere, Gestation and Phyllotaxis retain named operators. Geometry, scale, units and thermal/energy response require explicit laws. The example polynomials and blends are not certified as exact distance fields.
Finite graph; coupling; flow limit	Optional finite relational profile with declared node/edge budgets, weights, permissible transitions and throughput. Neither a graph nor small-world structure is compulsory in TOM.
Kiki UU	An open routing operator with declared closure, permitted visits and any non-intersection constraint. A result requires a feasibility check. A minimum-route claim additionally needs an objective and an optimality argument; loop closure alone is not that argument.
Observation and readout	A declared input channel supplies data, units, availability, calibration and uncertainty. Derived coordinates require an identifiable observation model. A detected zero means no admitted signal under that model, not a declaration of death or nonexistence.

[B pp. 1–10, 15, 26–28, 35–39; U]

Precedence is preserved locally, not imposed globally

B's morphological profile may retain

$$\text{PyramidSide} \prec \text{Cone} \prec \text{Sphere} \prec \text{Gestation} \prec \text{Phyllotaxis}(\phi) \prec \text{Klein}.$$

This orders that profile's declared dependencies. It neither replaces TOM's defining expression nor chooses intersection, union or smooth blending without a composition law. "SDF" retains A's definition-level signed-field meaning unless an application separately establishes a distance meaning. [A pp. 2, 5; B p. 9; U]

Required application declaration

Each bound profile supplies input/provenance, exact representation, ordered law and parameters, prediction, falsifier and reproducible evidence. Use distinct symbols for tick n , physical elapsed time t , temperature ϑ and coupling weight ω ; do not overload T or W across them. Start with synthetic data, then explicitly authorized measured inputs. [A p. 7; U]

Not established by these PDFs. Instantaneous global TSP optimization; log-free recovery of arbitrary past or future states; physical arrest from a sign conflict; universal two-hertz identity; remote biological inference; and automatic privacy, security or unclonability. These claims in B supply neither the required application laws nor validating evidence. They are not merged as proven results. No institutional affiliation or patent status is asserted. [A pp. 7, 9; B pp. 13–14, 18–34, 37–39]

06 / DELIVERY STATUS

Acceptance criteria and source traceability

Minimum conformance tests [U]

Test	Required outcome
Source preservation	The defining expression, ordered operands, law identities and retained evidence survive zero readouts, replay and cache reuse.
Exact constants	Verify $\phi^2 = 1 - \phi$, $\phi(1 + \phi) = 1$, the pair multiplication rule and the separate legacy basis. Never substitute an approximate logarithm.
Open-law handling	Every role returns a typed term or explicit status. Missing laws stay symbolic or block the concrete result that requires them.
Temporal admission	Reject retrograde/same-tick publication, future evidence and unsatisfied guards. A historical read cannot modify the prefix.
Atomic capacity	Force exhaustion in each budget. No failed step publishes a partial term or erases earlier evidence.
Gate and replay	$g = 0$ cannot publish through the active gate; $g = 1$ cannot bypass checks. Out-of-range history returns a status, not a fabricated state.
Adapter equivalence	Cached and uncached results agree exactly for the same complete inputs. Each observation or optimization claim has its own falsifiable test.

What still needs a supplied law

Neither attachment prints the complete numerical C recurrence, its initial state and the full Arch scalar law/domain needed for a closed numerical run. U1 therefore retains these as bound-or-symbolic contracts rather than inventing dynamics. A runnable numerical profile must also declare capacities, evidence encoding, temporal units and any observation or geometry law it uses. [A pp. 4–8]

A reports implementation files, CPU/GPU agreement and compatibility tests. Those files, test vectors, the parent v0.7 PDF and the complete TC1–TC5 texts were not supplied with these two attachments. Their reported results are not new validation of U1, especially after the explicit phi reconciliation. [A pp. 1–2, 8–10]

Source key

[A] TOM / v0.7.2 – *Universal SDF-Operator Profile*, supplied as TOM_V0_7_2_Formalization(1).pdf, 10 pages, dated 19 September 2026. Core authority here: closure and fidelity (pp. 1–2), field record and registry (pp. 3–5), transaction (p. 6), application boundary (p. 7), implementation claims and limits (pp. 8–10).

[B] The supplied 39-page exploratory dialogue beginning “*egg battery sturgeon farm gestation cycle lower phi ratio phyllotaxis*”. Relevant blocks: field operators and gate (pp. 1–12); coordinate-free transitions (pp. 14–17); finite lookup and memory (pp. 20–26); consolidated operators (pp. 35–36); just-in-time readout (pp. 37–39). Generated dialogue assertions are source material, not independent evidence.

[U] The reconciliation and design choices made in this document. They are a proposed merged specification, not a claim that either original already contained these resolutions.

Consolidated result. Exact signed-field closure with forward-only, provenance-preserving publication; one independent binary admission gate; bounded resources; and disposable, demand-driven application readouts. Topology, geometry, routing and physical sensing remain explicit, testable profiles.